

Smart Fire Safety Early Warning System of Large Complex Building based on Data Collection and Processing

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Abstract—In view of the current situation that the electrical design and intelligent special design of some projects do not overlap each other and the docking design is out of touch, combined with the electrical intelligent design practice of a large commercial complex, the hidden dangers of electrical fire safety in heavy high-rise buildings cannot be ignored, especially In high-rise complex buildings, some high-power operating equipment in the electrical equipment will directly cause the temperature of the wires to rise, which is prone to fires. A reasonable electrical fire protection design can prevent the occurrence and spread of fires. This paper mainly studies the electrical intelligent fire safety early warning system of large-scale complex buildings and its application scenarios

Keywords—Large complex building, Electrical intelligence, Smart fire, Fire safety warning

I. INTRODUCTION

With the development of the market economy, the city's high-rise buildings have sprung up like bamboo shoots after a rain, and they are gradually becoming integrated, high-rise, and modern. In modern urban life, in order to improve the efficiency of life and work, people hope to quickly solve the needs of life and work in the same space. The high-rise complex building is a landmark building of a city. It brings together department stores, entertainment, supermarkets, apartments, residences, restaurants and other functions. It has large spaces, a large number of vertical transportation systems, and various high-tech facilities are set up in a centralized manner. Typical characteristics of modern buildings. The huge volume of the building, complex functional structure, and a large number of various types of electrical equipment make the architectural design more complicated. Therefore, building modernization has become an objective requirement of my country's socialist modernization, and building fire-fighting modernization has also become an integral part of building modernization [1-7].

The diversified integration of functions and the nature of use of high-rise complexes determine that when a fire occurs, it has its own significant characteristics compared with general building fires. The characteristics of multiple types of fires are at the same time and space. People gather and the space is complicated. Evacuation and rescue are quite difficult. Higher requirements are put forward for the indication of the evacuation lighting system. There are many decorative materials, large fire loads, toxic smoke and high-temperature

smoke. It is easy to cause casualties, and higher requirements are put forward on the materials used for cables and the wiring system; in addition, the complex often uses a variety of lighting fixtures, high-power air-conditioning equipment, computer audio control equipment and other electrical equipment, and the wiring is complicated and consumes the battery is large. For fixed fire locations such as stoves and boiler rooms, once liquid and gaseous fuel leaks and meets open flames or accidental use of fire, it is easy to cause fires. This requires that electrical fire protection design needs to analyze the situation of each site in depth, and the lighting and power distribution lines Corresponding design should be made for the selection and laying of the fire alarm and the linkage configuration of the gas alarm system[8-12].

The electrical and intelligent design process nodes mainly include: plan review, first edition construction drawing review, and second edition construction drawing review. The design of each stage of the intelligent special project usually starts after the review node of the civil engineering and electrical phases is passed, and the capital is raised in conjunction with the civil engineering, electromechanical, etc. The time node lags behind the building electrical professional. Combining the practice of this project, the integration of electrical and intelligent design requires the advancement of intelligent special process nodes at various stages. The electric capacity, the location and size of the weak current wells, the specifications of the main line bridge, the direction, etc. shall be designed in advance. After the integration of the building electrical designer, the capital plan or construction profession will be raised, and the integrated design will be implanted in the building from the source of the plan to avoid the later period adjustment [13-18].

A fire in a complex building can cause huge economic losses. The complex building volume is huge, and the main structure of the building and the later decoration and construction cost are relatively high, basically in the tens of millions or even more than 100 million. If a fire occurs and the rescue is not done in time, huge economic losses will occur, which will directly exceed the standards defined by my country for a catastrophic fire, which will affect local economic construction and the safety of people's lives and property; there will be heavy casualties. There are a large number of people in the complex building, the number of combustible and combustible materials is huge, and there are many hidden safety hazards. The fire spreads rapidly, and the

smoke and poisonous gas produced by the fire affect the safe evacuation of people. At the same time, the building structure has a limited fire resistance time. Once this time is exceeded, the structure may collapse due to overloading, causing casualties and affecting social stability. When a fire occurs in a complex building, it will cause various degrees of economic losses and casualties, which will affect stability and not be conducive to economic development. This requires electrical fire-fighting designers to fully understand the situation of complex buildings, rationally arrange and lay lighting lines, power distribution lines, fire alarm systems, gas alarm systems, etc., and set up linkage configurations reasonably to improve the role of the fire protection system. But nowadays, there are many units in the design and construction of complex buildings, and the decoration and design are usually not in a period of time. This makes the content of the building have many hidden safety hazards. There are still many problems in deepening and coordinating the design that require research [19-24].

II. THE PROPOSED METHODOLOGY

A. Large Complex Building Fire Protection

A high-rise complex building is located in the capital city of the western province, with a total construction area of approximately 3×104 square meters and a total building height of 89.6m, including two underground floors and 18 floors above ground. The 2 basement floors are the parking lot, the basement 1-the 5 floors above the ground are commercial floors, the 6th-16 floors are hotel rooms, and the 17-18 floors are office buildings. The fire resistance grade is I grade, the roof is waterproof, and the basement is waterproof, and the service life of the building is 50 years. This is a typical commercial complex building with two power sources on the first floor underground, both of which are 10kV: high-voltage power distribution room and substation room.

$$f(I; \theta) = \{m_i\}_{i=1}^J, m \in [0, 1]^{h \times w} \quad (1)$$

$$x_s^{input} = g_s^{ctx}(x_{(s-1)}^{ctx}, x_s^v, \gamma_s) \quad (2)$$

The reliability of the fire-fighting load power supply is ensured by setting up diesel power stations. When the 10kV power supply fails, the diesel power station generator set will start and provide power to support fire-fighting electrical equipment within 15s. When designing the fire protection electrical design of high-rise complex buildings, the overall situation of the building and the fire protection facilities should be considered to ensure the scientific rationality of the design plan. The combination scheme of fire alarm and linkage bus is more flexible, but the reliability is relatively low. Therefore, the fire alarm and fire linkage bus are selected to be independent in the electrical fire protection design to ensure reliability. Install the system according to the manufacturer's requirements, and learn more about the relevant data map before installation. Use conduits and cables to connect the devices in the system. Except for special requirements, the cable specifications are $\geq 16\text{AWG}$, and the

"fire protection system" label is affixed to the outside of the junction box.

The project is located in Dongcheng District, Dongguan City, Guangdong Province, east of Dongguan Avenue, north of Hongfu Road, south of Zanhua Road, and west of Planning Branch Road. The land area of the project is 104,870 m², the plot ratio is 6.13, the capacity-rated building area is 643,000 m², and the total building area is 1.035 million m². The overall plan is 5 high-rise buildings. Building No. 1 (No. T1), office type, building height 190.5m; Building No. 2 (No. T2), office type, building height 392.2m; Building No. 3 (No. T3), low area for office, high area for apartment type, The building height is 219.15m; Building No. 4 (No. T4) is a five-star hotel with a building height of 100m; Building No. 5 (No. T5) contains commercial podiums with 1 to 7 floors above the ground and a commercial building with a height of 82.5m tower. The basement of the project has 4 floors, the semi-basement is for garages and equipment rooms; the 1-2 floors are for commercial, bus stations, garages and equipment rooms; the 3 to 4 floors for underground are garages, equipment rooms, and air defenses.

B. Electrical Intelligent Fire Protection System

When point-type smoke detectors are installed on a ceiling with beams, they meet the following requirements:

a. When the height of the beam protruding from the ceiling is less than 200mm, the influence of the beam on the protection area of the detector is not considered.

b. When the height of the beam protruding from the ceiling is 200mm~600mm, determine the influence of the beam on the protection area of the detector and the number of areas between the beams that a detector can protect according to the GB50116-2013 specification.

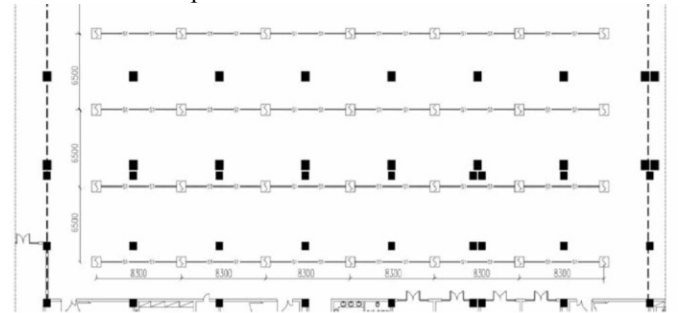


Fig.1. Emergency Rehabilitation Management System

When the height of the beam protruding from the ceiling exceeds 600mm, at least one detector shall be installed in each area between beams separated by the beam. When point detectors are installed on the ceiling of the inner walkway with a width less than 3m, they shall be arranged in the center. The installation distance of temperature-sensing fire detectors shall not exceed 10m, and the installation distance of smoke-sensing fire detectors shall not exceed 15m; the distance between the detectors and the end wall shall not exceed 7m. The horizontal distance between the point detector and the edge of the wall and beam shall not be less than 0.5m. When the room is separated by equipment or partitions, and the distance from the top to the ceiling beam is less than 5% of the

clear height of the room (such as basement garages and equipment rooms), at least one point detector should be installed in each separated part. The point detector should be installed close to the return air outlet, and the horizontal distance to the air outlet of the air conditioner should be greater than 1.5m. When the hole of the porous air supply ceiling and the point detector are arranged on the same plane, the horizontal distance should be greater than 0.5m. The location of the manual fire alarm button is generally on the wall of the evacuation passage or entrance. Each fire compartment has at least one manual fire alarm button, from the farthest position in a fire compartment to the nearest manual fire alarm button. The distance is less than 30m.

The manual fire alarm button is set in an obvious and easy-to-operate position. When the wall-mounted installation is used, the height from the bottom edge to the ground is 1.4m, which is usually a red panel, which is more eye-catching. A regional fire monitor is set in the area under the jurisdiction of each alarm loop, which is also called a fire monitor for short; shopping malls, hotels, supermarkets, garages, restaurants.

C. Fire Safety Early Warning System

The intelligent computer room of this project includes: a large commercial consumer control monitoring center computer room (about 130m²) on the first floor, a Jinjie consumer control monitoring room (about 40m²); an operator access computer room (about 50m², it is shared for the entire project); there is a commercial management information computer room (about 60m²) on the third floor. In addition to the passenger flow statistics analysis system, wireless WiFi system, commercial management office system and POS cash register system, the host is located in the three-story commercial management information computer room, other intelligent integrated systems, information application systems, commercial management application systems, public safety systems, and buildings. The equipment management system and automatic fire alarm system and other electrical fire control system hosts are equally divided into types and located in the consumer control monitoring room on the first floor. The consumer control monitoring room is the "brain" of the smart building, and its importance is self-evident.

$$V = \arg \min \sum_{i=1}^k |x_j - u_i|^2 \quad (3)$$

$$T = \overline{X_1} - \overline{X_2} \quad (4)$$

Because the fire control monitoring room covers the automatic fire alarm system and most of the intelligent systems, it also involves the docking of the computer room fire protection power distribution and lightning protection grounding systems, which is a place where electrical and intelligent professional design have many intersections. In the early stage of this project, it is clear that the main control major of the control room is an intelligent special design. First, the intelligent major divides the computer room into the operation area and the equipment area, and divides the fire protection system and other intelligent systems according to the specifications, and then sets the TV The walls, consoles, cabinets, battery cabinets, bridges and other equipment are

arranged, and the electrical design is to rationally arrange the fire cabinets, wall-mounted hosts, equipotential boxes and dual power boxes, etc., and finally the intelligent design for overall integrated design, To meet the requirements of a safe, reliable, efficient and clean computer room.

III. EXPERIMENT

The effect of the fire safety early warning system of traditional buildings is shown.

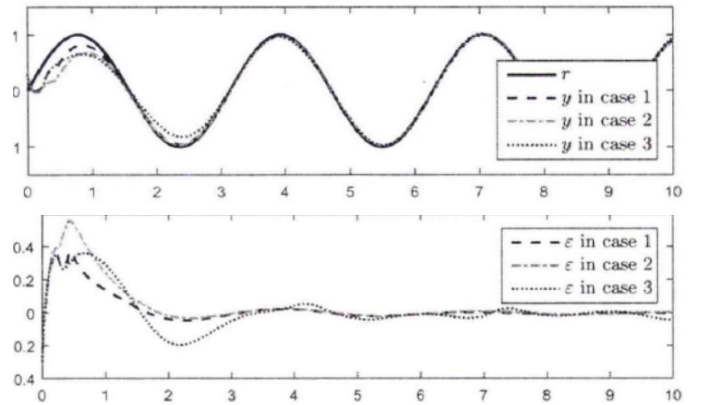


Fig. 2. Safety early warning system effect

The electrical intelligent fire safety system of a large complex building is shown in the figure.

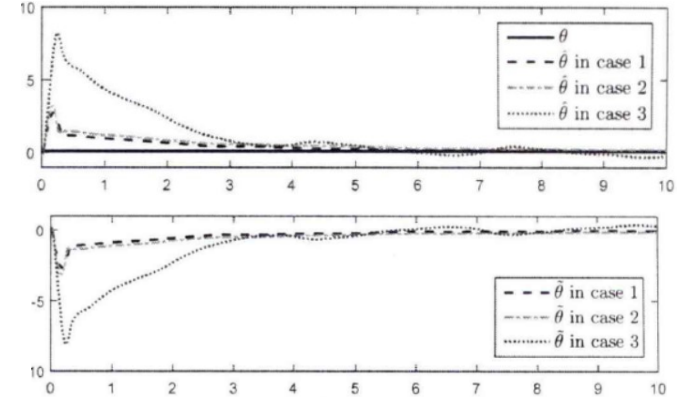


Fig. 3. Electrical intelligent fire safety system

The effect of the electrical intelligent fire safety early warning system for large complex buildings is shown in the figure.

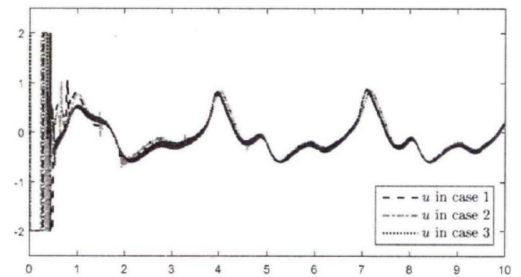


Fig. 4. Electrical safety early warning system effect

IV. CONCLUSION

First to achieve integrated design from electrical and intelligent, and at the same time actively cooperate with civil engineering, plumbing and various external special design professions, make overall plans, and truly integrate the concept of integration. Throughout the design, improve the completion and satisfaction of each smart building. This article combines the design practice of a large-scale commercial complex project to complete the electrical intelligent fire safety early warning system from the whole process of preliminary plan planning, construction drawing design and later operation and maintenance.

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